Your Success is our Win

Ex # 4.1

(1) Find the n^{th} term (rule of formation) of each of the following sequences.

(i) $2, 4, 6, \ldots$

Sol

$a_1 = 2 = 2(1)$

$a_2 = 4 = 2(2)$

$a_3 = 6 = 2(3)$

From above pattern we deduced that $a_n = 2n$.

(2) $1^2, 2^2, 3^2$

Sol

Here,

$a_1 = 1^2$

$a_2 = 2^2$

$a_3 = 3^2$

From above pattern we deduced that $a_n = n^2$

(3) $3, 9, 27$

Sol

$3, 9, 27$

$a_1 = 3 = 3^1$

$a_2 = 9 = 3^2$

$a_3 = 27 = 3^3$

From above pattern we deduced that $a_n = 3^n$.

(iv) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}$

Sol

Here,

$a_1 = 1/2 = 1/1+1$

$a_2 = 2/3 = 2/2+1$

$a_3 = 3/4 = 3/3+1$

$a_4 = 4/5 = 4/4+1$

From above pattern we deduced that $a_n = n/n+1$

(v) $\frac{1}{1}, \frac{-1}{3}, \frac{1}{9}, \frac{-1}{27}$

Sol

Here,

$a_1 = 1 = 1/(-3)^{1-1}$

$a_2 = -1/3 = 1/(-3)^{2-1}$

$a_3 = 1/9 = 1/(-3)^{3-1}$

$a_4 = -1/27 = 1/(-3)^{4-1}$

From above pattern we deduced that $a_n = 1/(-3)^{n-1}$

or

$a_n = (1/-3)^{n-1}$

(vi) 1, 8, 27, 64

Sol

Here,

$a_1 = 1 = 1^3$

$a_2 = 8 = 2^3$

$a_3 = 27 = 3^3$

$a_4 = 64 = 4^3$

From above pattern we deduced that $a_n = n^3$

(2) Find the first five terms of sequences with given general term -

(i) $2n/3\,(n+1)$

Sol

For $n=1$,

$\Rightarrow 2(1)/3\,(1+1) = 4/3$

For $n=2$,

$\Rightarrow 2(2)/3\,(2+1) = 4$

For $n=3$,

$\Rightarrow \frac{2(3)}{3}(3+1) = 8$

For $n=4$,

$\Rightarrow \frac{2(4)}{3}(4+1) = \frac{40}{3}$

For $n=5$,

$\Rightarrow \frac{2(5)}{3}(5+1) = 20$

Thus, the required sequence is $\frac{4}{3}, 4, 8, \frac{40}{3}, 20, \ldots$

A

(ii) $(-1)^{n+1} \cdot 3^{n-1}$

Sol

For $n=1$,

$\Rightarrow (-1)^{1+1} \cdot 3^{1-1} = 1$

For $n=2$,

$\Rightarrow (-1)^{2+1} \cdot 3^{2-1} = -3$

For $n=3$,

$\Rightarrow (-1)^{3+1} \cdot 3^{3-1} = 9$

For $n=4$,

$\Rightarrow (-1)^{4+1} \cdot 3^{4-1} = -27$

For $n=5$,

$\Rightarrow (-1)^{5+1} \cdot 3^{5-1} = 81$

Thus the required sequence is $(1, -3, 9, -27, 81, \ldots)$

A

(iii) $\frac{1}{4}n^2(n+1)^2$

Sol

For $n=1$,

$\Rightarrow \frac{1}{4}(1)^2(1+1)^2 = 1$

For $n=2$,

$\Rightarrow \frac{1}{4}(2)^2(2+1)^2 = 9$

For $n=3$,

$\Rightarrow \frac{1}{4}(3)^2(3+1)^2 = 36$

For $n=4$,

$\Rightarrow \frac{1}{4}(4)^2(4+1)^2 = 100$

For $n=5$,

$\Rightarrow \frac{1}{4}(5)^2(5+1)^2 = 225$

Thus, the required sequence is $1, 9, 36, 100, 225, \ldots$

Ans

(iv) $n/3n+1$

Sol

For $n=1$,

$\Rightarrow \frac{1}{3(1)+1} = \frac{1}{4}$

For $n=2$,

$\Rightarrow \frac{2}{3(2)+1} = \frac{2}{7}$

For $n=3$,

$\Rightarrow \frac{3}{3(3)+1} = \frac{3}{10}$

For $n=4$,

$\Rightarrow \frac{4}{3(4)+1} = \frac{4}{13}$

$\Rightarrow$ for $n=5$,

$\Rightarrow \frac{5}{3(5)+1} = \frac{5}{16}$

Thus the required sequence is $\frac{1}{4}, \frac{2}{7}, \frac{3}{10}, \frac{4}{13}, \frac{5}{16}, \ldots$

Ans

(v) $\frac{1}{6}n(n+1)(n+2)$

Sol

For $n=1$,

$\Rightarrow \frac{1}{6}(1)(1+1)(1+2) = 1$

For $n=2$,

$\Rightarrow \frac{1}{6}(2)(2+1)(2+2) = 4$

For $n=3$,

$\Rightarrow \frac{1}{6}(3)(3+1)(3+2) = 10$

For $n=4$,

$\Rightarrow \frac{1}{6}(4)(4+1)(4+2) = 20$

For $n=5$,

$\Rightarrow \frac{1}{6}(5)(5+1)(5+2) = 35$

Thus the required sequence is $1, 4, 10, 20, 35, \ldots$

Ans.

(vi) $\frac{1}{6} n(n+1)(2n+7)$

Sol.

For $n=1$,

$\Rightarrow \frac{1}{6}(1)(1+1)\{2(1)+7\} = 3$

For $n=2$,

$\Rightarrow \frac{1}{6}(2)(2+1)\{2(2)+7\} = 11$

For $n=3$,

$\Rightarrow \frac{1}{6}(3)(3+1)\{2(3)+7\} = 26$

For $n=4$,

$\Rightarrow \frac{1}{6}(4)(4+1)\{2(4)+7\} = 50$

For $n=5$,

$\Rightarrow \frac{1}{6}(5)(5+1)\{2(5)+7\} = 85$

Thus the required sequence is $3, 11, 26, 50, 85, \ldots$

Ans.

(vii) $a_{n+1} = 6 + a_n$, $a_1 = 2$

Sol.

For $n=1$,

$\Rightarrow a_{1+1} = 6 + a_1$

$\Rightarrow a_2 = 6 + 2 = 8$

For $n=2$,

$\Rightarrow a_{2+1} = 6 + a_2$

$\Rightarrow a_3 = 6 + 8 = 14$

For $n=3$,

$\Rightarrow a_{3+1} = 6 + a_3$

$\Rightarrow a_4 = 6 + 14 = 20$

For $n=4$,

$\Rightarrow a_{4+1} = 6 + a_4$

$\Rightarrow a_5 = 6 + 20 = 26$

Thus, the required sequence is $2, 8, 14, 20, 26, \ldots$

Ans.

(viii) $a_n = n/2\ \cancel{(n+1)}\ a_{n+1}$, $a_1 = 1$

Sol.

For $n=1$,

$\Rightarrow a_1 = 1/2\,(a_1+1)$

$\Rightarrow 1 = 1/2\,(a_2)$

$\Rightarrow a_2 = 2$

For $n=2$,

$\Rightarrow a_2 = \frac{2}{2}(a_{2+1})$

$\Rightarrow 2 = \cancel{2}/\cancel{2}(a_3)$

$\Rightarrow a_3 = 2$

For $n=3$,

$\Rightarrow a_3 = \frac{3}{2}(a_{3+1})$

$\Rightarrow 2 = 3/2(a_4)$

$\Rightarrow a_4 = 4/3$

For $n=4$,

$\Rightarrow a_4 = \frac{4}{2}(a_{4+1})$

$\Rightarrow 4/3 = 4/2(a_5)$

$\Rightarrow a_5 = 8/12 = 2/3$

Thus the required sequence is $1, 2, 2, 4/3, 2/3, \ldots$

Ans.

(3) Find the sequence by using $T_{n+1} = (n+1)T_n$, where $T_1 = 2$

Sol

For $n=1$,

$\Rightarrow T_{1+1} = (1+1)T_1$

$\Rightarrow T_2 = 2(2) = 4$

For $n=2$,

$\Rightarrow T_{2+1} = (2+1)T_2$

$\Rightarrow T_3 = 3(4) = 12$

For $n=3$,

$\Rightarrow T_{3+1} = (3+1)T_3$

$\Rightarrow T_4 = (4)(12) = 48$

For $n=4$,

$\Rightarrow T_{4+1} = (4+1)T_4$

$\Rightarrow T_5 = (5)(48)$

$\Rightarrow T_5 = 240$

Thus the required sequence is $2, 4, 12, 48, 240, \ldots$

Ans.

(4) Find the value of n^{th} term of triangular sequence when $n = 7, 9, 12$ and 16.

Sol

As we know that,

$$t_n = \frac{n(n+1)}{2}$$

For $n=7$,

$\Rightarrow t_7 = \frac{7(7+1)}{2}$

$\Rightarrow t_7 = 28$

For $n=9$

$\Rightarrow t_9 = \frac{9(9+1)}{2}$

$\Rightarrow t_9 = 45$

For $n=12$

$\Rightarrow t_{12} = \frac{12(12+1)}{2}$

$\Rightarrow t_{12} = 78$

For $n=16$

$\Rightarrow t_{16} = \frac{16(16+1)}{2}$

$\Rightarrow t_{16} = \frac{\overset{8}{\cancel{16}}(17)}{\underset{1}{\cancel{2}}}$

$\Rightarrow t_{16} = 8(17)$

$\Rightarrow t_{16} = 136$

Thus the required sequence is 28, 45, 78, 136.

(5) Find the first five terms of sequence with general term: $a_n = \frac{(n+1)!}{2}$

Sol:

For $n=1$,

$\Rightarrow a_1 = \frac{(1+1)!}{2}$

$\Rightarrow a_1 = \frac{2!}{2} = 1$

For $n=2$,

$\Rightarrow a_2 = \frac{(2+1)!}{2}$

$\Rightarrow a_2 = \frac{3!}{2} = \frac{6}{2} = 3$

For $n=3$,

$\Rightarrow a_3 = \frac{(3+1)!}{2} = \frac{4!}{2} = \frac{24}{2} = 12$

For $n=4$,

$\Rightarrow a_4 = \frac{(4+1)!}{2} = \frac{5!}{2} = \frac{120}{2} = 60$

For $n=5$,

$\Rightarrow a_5 = \frac{(5+1)!}{2} = \frac{6!}{2} = \frac{720}{2} = 360$

Thus the required sequence is 1, 3, 12, 60, 360

Find the Pascal's sequence when $n = 5, 6, 7$ and 8.

Sol:

For find Pascal sequence we use formula

$\frac{n!}{r!(n-r)!}$, $r \leq n$

By using formula of pascal sequence when $n = 5$ and $r = 0, 1, 2, 3, 4, 5$.

At $r = 0$; $\frac{5!}{(5-0)!\,0!} = \frac{5!}{5!} = 1$

At $r = 1$, $\frac{5!}{(5-1)!\,1!} = \frac{5!}{4!} = \frac{5 \times 4!}{4!} = 5$

At $r = 2$, $\frac{5!}{(5-2)!\,2!} = \frac{5 \times 4 \times \cancel{3!}}{\cancel{3!}\,2!} = \frac{20}{2} = 10$

At $r = 3$, $\frac{5!}{(5-3)!\,3!} = \frac{5 \times 4 \times \cancel{3!}}{2 \times \cancel{3!}} = 10$

At $r = 4$, $\frac{5!}{(5-4)!\,4!} = \frac{5 \times 4!}{4!} = 5$

At $r = 5$, $\frac{5!}{(5-5)!\,5!} = \frac{5!}{5!} = 1$

Thus pascal sequence when $n = 5$ is $1, 5, 10, 10, 5, 1$.

By using formula of Pascal sequence when $n = 6$ and $r = 0, 1, 2, 3, 4, 5, 6$.

At $r = 0$, $\frac{6!}{(6-0)!0!} = \frac{6!}{6!} = 1$

At $r = 1$, $\frac{6!}{(6-1)!1!} = \frac{6!}{5!} = \frac{6\times\cancel{5!}}{\cancel{5!}} = 6$

At $r = 2$, $\frac{6!}{(6-2)!2!} = \frac{6!}{4!\times 2} = \frac{\overset{3}{\cancel{6}}\times 5\times\cancel{4!}}{\cancel{4!}\times\cancel{2}} = 15$

At $r = 3$, $\frac{6!}{(6-3)!3!} = \frac{6!}{3!\times 6} = \frac{\cancel{6}\times 5\times 4\times\cancel{3!}}{\cancel{3!}\times\cancel{6}} = 20$

At $r = 4$, $\frac{6!}{(6-4)!4!} = \frac{6!}{2\times 4!} = \frac{6\times 5\times 4!}{2\times 4!} = 15$

At $r = 5$, $\frac{6!}{(6-5)!5!} = \frac{6\times\cancel{5!}}{\cancel{5!}} = 6$

At $r = 6$, $\frac{6!}{(6-6)!6!} = \frac{\cancel{6!}}{\cancel{6!}} = 1$

Thus Pascale sequence when $n = 6$ is 1, 6, 15, 20, 15, 6, 1

By using formula of Pascal sequence when $n = 7$ and $r = 0, 1, 2, 3, 4, 5, 6, 7$

At $r = 0$, $\frac{7!}{(7-0)!0!} = \frac{7!}{7!} = 1$

At $r = 1$, $\frac{7!}{(7-1)!1!} = \frac{7\times\cancel{6!}}{\cancel{6!}\times 1} = 7$

At $r = 2$, $\frac{7!}{(7-2)!2!} = \frac{7\times\overset{3}{\cancel{6}}\times\cancel{5!}}{\cancel{5!}\times\cancel{2}} = 21$

At $r=3$, $\frac{7!}{(7-3)!3!} = \frac{7\times6\times5\times4!}{4!\times6} = 35$

At $r=4$, $\frac{7!}{(7-4)!4!} = \frac{7\times6\times5\times4!}{3!\times4!} = \frac{7\times6\times5}{6} = 35$

At $r=5$, $\frac{7!}{(7-5)!5!} = \frac{7\times6\times5!}{2!\times5!} = \frac{42}{2} = 21$

At $r=6$, $\frac{7!}{(7-6)!6!} = \frac{7\times6!}{1\times6!} = 7$

At $r=7$, $\frac{7!}{(7-7)!7!} = \frac{7!}{7!} = 1$

Thus Pascal sequence when $n=7$ is 1, 7, 21, 35, 35, 21, 7, 1.

By using Pascal formula when $n=8$ and $r=0$, 1, 2, 3, 4, 5, 6, 7 and 8-

At $r=0$, $\frac{8!}{(8-0)!0!} = \frac{8!}{8!} = 1$

At $r=1$, $\frac{8!}{(8-1)!1!} = \frac{8\times7!}{7!} = 8$

At $r=2$, $\frac{8!}{(8-2)!2!} = \frac{{}^{4}8\times7\times6!}{6!\times2!} = 28$

At $r=3$, $\frac{8!}{(8-3)!3!} = \frac{8\times7\times6\times5!}{5!\times6} = 56$

At $r=4$, $\frac{8!}{(8-4)!4!} = \frac{8\times7\times6\times5\times4!}{4!\times24} = 70$

At $r = 5$, $\frac{8!}{(8-5)!5!} = \frac{8 \times 7 \times 6 \times \cancel{5!}}{3! \times \cancel{5!}} = 56$

At $r = 6$, $\frac{8!}{(8-6)!6!} = \frac{8 \times 7 \times \cancel{6!}}{2! \times \cancel{6!}} = \frac{56}{2} = 28$

At $r = 7$, $\frac{8!}{(8-7)!7!} = \frac{8 \times 7!}{7!} = 8$

At $r = 8$, $\frac{8!}{(8-8)!8!} = \frac{8!}{8!} = 1$

Thus Pascal sequence when $n = 8$ is 1, 8, 28, 56, 70, 56, 28, 8, 1

Ans